Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

Welcome | mongosh: mongo

>_MONGOSH

```
> use("library_db")
< switched to db library_db
> db.members.insertMany([
    { member_id: 1, name: "Aarav", city: "Hyderabad", membership_type: "Gold" },
    { member_id: 2, name: "Priya", city: "Bangalore", membership_type: "Silver" },
    { member_id: 3, name: "Rahul", city: "Mumbai", membership_type: "Gold" },
    { member_id: 4, name: "Sneha", city: "Delhi", membership_type: "Silver" },
    { member_id: 5, name: "Kiran", city: "Hyderabad", membership_type: "Gold" }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69df0997b5a535524a862e2e'),
      '1': ObjectId('69df0997b5a535524a862e2f'),
      '2': ObjectId('69df0997b5a535524a862e30'),
      '3': ObjectId('69df0997b5a535524a862e31'),
      '4': ObjectId('69df0997b5a535524a862e32')
    }
  }
> db.books.insertMany([
    { book_id: 101, title: "MongoDB Basics", category: "Database", author: "John Smith", price: 500 },
    { book_id: 102, title: "Python Fundamentals", category: "Programming", author: "Alice Brown", price: 650 },
    { book_id: 103, title: "Data Engineering Intro", category: "Data", author: "Mark Lee", price: 800 },
    { book_id: 104, title: "SQL for Beginners", category: "Database", author: "David Miller", price: 450 },
    { book_id: 105, title: "Machine Learning Start", category: "AI", author: "Sara Khan", price: 900 }
```

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  - library_db
    - members
  - local
  - retail-db
  - retail_db

Welcome | mongosh: mongo

>_MONGOSH

```
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69df09cfb5a535524a862e38'),
    '1': ObjectId('69df09cfb5a535524a862e39'),
    '2': ObjectId('69df09cfb5a535524a862e3a'),
    '3': ObjectId('69df09cfb5a535524a862e3b'),
    '4': ObjectId('69df09cfb5a535524a862e3c'),
    '5': ObjectId('69df09cfb5a535524a862e3d')
  }
}
> db.members.find()
< {
  _id: ObjectId('69df0997b5a535524a862e2e'),
  member_id: 1,
  name: 'Aarav',
  city: 'Hyderabad',
  membership_type: 'Gold'
}
{
  _id: ObjectId('69df0997b5a535524a862e2f'),
  member_id: 2,
  name: 'Priya',
  city: 'Bangalore',
  membership_type: 'Silver'
```

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
    member_id: 5,
    name: 'Kiran',
    city: 'Hyderabad',
    membership_type: 'Gold'
  }
> db.books.find()
< {
    _id: ObjectId('69df09c2b5a535524a862e33'),
    book_id: 101,
    title: 'MongoDB Basics',
    category: 'Database',
    author: 'John Smith',
    price: 500
  }
  {
    _id: ObjectId('69df09c2b5a535524a862e34'),
    book_id: 102,
    title: 'Python Fundamentals',
    category: 'Programming',
    author: 'Alice Brown',
    price: 650
  }
  {
    _id: ObjectId('69df09c2b5a535524a862e35'),
```

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
> db.borrowings.find()
< {
    _id: ObjectId('69df09cfb5a535524a862e38'),
    borrow_id: 1001,
    member_id: 1,
    book_id: 101,
    days_borrowed: 5,
    borrow_date: '2026-03-01'
  }
  {
    _id: ObjectId('69df09cfb5a535524a862e39'),
    borrow_id: 1002,
    member_id: 2,
    book_id: 102,
    days_borrowed: 3,
    borrow_date: '2026-03-02'
  }
  {
    _id: ObjectId('69df09cfb5a535524a862e3a'),
    borrow_id: 1003,
    member_id: 1,
    book_id: 103,
    days_borrowed: 7,
    borrow_date: '2026-03-03'
```

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

```
}
> db.members.find({ city: "Hyderabad" })
< {
    _id: ObjectId('69df0997b5a535524a862e2e'),
    member_id: 1,
    name: 'Aarav',
    city: 'Hyderabad',
    membership_type: 'Gold'
  }
  {
    _id: ObjectId('69df0997b5a535524a862e32'),
    member_id: 5,
    name: 'Kiran',
    city: 'Hyderabad',
    membership_type: 'Gold'
  }
> db.books.find({ category: "Database" })
< {
    _id: ObjectId('69df09c2b5a535524a862e33'),
    book_id: 101,
    title: 'MongoDB Basics',
    category: 'Database',
    author: 'John Smith',
    price: 500
  }
```

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Data Modeling

ONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
> db.books.find({ price: { $gt: 600 } })
< {
    _id: ObjectId('69df09c2b5a535524a862e34'),
    book_id: 102,
    title: 'Python Fundamentals',
    category: 'Programming',
    author: 'Alice Brown',
    price: 650
  }
  {
    _id: ObjectId('69df09c2b5a535524a862e35'),
    book_id: 103,
    title: 'Data Engineering Intro',
    category: 'Data',
    author: 'Mark Lee',
    price: 800
  }
  {
    _id: ObjectId('69df09c2b5a535524a862e37'),
    book_id: 105,
    title: 'Machine Learning Start',
    category: 'AI',
    author: 'Sara Khan',
    price: 900
```

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Data Modeling

ONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
> db.borrowings.find({ days_borrowed: { $gt: 5 } })
< {
    _id: ObjectId('69df09cfb5a535524a862e3a'),
    borrow_id: 1003,
    member_id: 1,
    book_id: 103,
    days_borrowed: 7,
    borrow_date: '2026-03-03'
  }
  {
    _id: ObjectId('69df09cfb5a535524a862e3c'),
    borrow_id: 1005,
    member_id: 5,
    book_id: 105,
    days_borrowed: 10,
    borrow_date: '2026-03-07'
  }
> db.books.find().sort({ price: -1 })
< {
    _id: ObjectId('69df09c2b5a535524a862e37'),
    book_id: 105,
    title: 'Machine Learning Start',
    category: 'AI',
    author: 'Sara Khan',
```

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My Queries

Data Modeling

ONNECTIONS (1)

Search connections

mongo

- HexaDB
- admin
- config
- library_db
  - members
- local
- retail-db
- retail_db

>_MONGOSH

```
> db.members.find().sort({ name: 1 })
< {
    _id: ObjectId('69df0997b5a535524a862e2e'),
    member_id: 1,
    name: 'Aarav',
    city: 'Hyderabad',
    membership_type: 'Gold'
  }
  {
    _id: ObjectId('69df0997b5a535524a862e32'),
    member_id: 5,
    name: 'Kiran',
    city: 'Hyderabad',
    membership_type: 'Gold'
  }
  {
    _id: ObjectId('69df0997b5a535524a862e2f'),
    member_id: 2,
    name: 'Priya',
    city: 'Bangalore',
    membership_type: 'Silver'
  }
  {
    _id: ObjectId('69df0997b5a535524a862e30'),
```

My Queries

Data Modeling

ONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
    city: 'Delhi',
    membership_type: 'Silver'
  }
> db.members.countDocuments()
< 5
> db.books.countDocuments()
< 5
> db.books.countDocuments({ category: "Database" })
< 2
> db.books.aggregate([
    { $group: { _id: null, avgPrice: { $avg: "$price" } } }
  ])
< {
    _id: null,
    avgPrice: 660
  }
> db.books.aggregate([
    { $group: { _id: null, maxPrice: { $max: "$price" } } }
  ])
< {
    _id: null,
    maxPrice: 900
  }
> db.books.aggregate([
```

My Queries

Data Modeling

ONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
  }
> db.borrowings.aggregate([
    {
      $lookup: {
        from: "books",
        localField: "book_id",
        foreignField: "book_id",
        as: "book_info"
      }
    }
  ])
< {
    _id: ObjectId('69df09cfb5a535524a862e38'),
    borrow_id: 1001,
    member_id: 1,
    book_id: 101,
    days_borrowed: 5,
    borrow_date: '2026-03-01',
    book_info: [
      {
        _id: ObjectId('69df09c2b5a535524a862e33'),
        book_id: 101,
        title: 'MongoDB Basics',
        category: 'Database',
        author: 'John Smith',
```

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My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
> db.borrowings.aggregate([
    {
      $lookup: {
        from: "members",
        localField: "member_id",
        foreignField: "member_id",
        as: "member"
      }
    },
    { $unwind: "$member" },
    {
      $lookup: {
        from: "books",
        localField: "book_id",
        foreignField: "book_id",
        as: "book"
      }
    },
    { $unwind: "$book" },
    {
      $project: {
        _id: 0,
        member_name: "$member.name",
        book_title: "$book.title"
      }
```

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My Queries

Data Modeling

NNECTIONS (1)

Search connections

mongo

- HexaDB
- admin
- config
- library_db
  - members
- local
- retail-db
- retail_db

>_MONGOSH

```
  }
  {
    member_name: 'Priya',
    book_title: 'Python Fundamentals'
  }
  {
    member_name: 'Aarav',
    book_title: 'Data Engineering Intro'
  }
  {
    member_name: 'Rahul',
    book_title: 'SQL for Beginners'
  }
  {
    member_name: 'Kiran',
    book_title: 'Machine Learning Start'
  }
  {
    member_name: 'Kiran',
    book_title: 'MongoDB Basics'
  }
> db.borrowings.aggregate([
  {
    $group: {
      _id: "$book_id"
```

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{} My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
      total_borrowed: "$total"
    }
  }
])
{
  book_title: 'SQL for Beginners',
  total_borrowed: 1
}
{
  book_title: 'Python Fundamentals',
  total_borrowed: 1
}
{
  book_title: 'MongoDB Basics',
  total_borrowed: 2
}
{
  book_title: 'Data Engineering Intro',
  total_borrowed: 1
}
{
  book_title: 'Machine Learning Start',
  total_borrowed: 1
}
```

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My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

>_MONGOSH

```
    _id: "A1",
    total: 1
  }
> db.borrowings.aggregate([
    {
      $group: {
        _id: "$member_id",
        count: { $sum: 1 }
      }
    },
    { $match: { count: { $gt: 1 } } }
  ])
< {
    _id: 5,
    count: 2
  }
  {
    _id: 1,
    count: 2
  }
> db.borrowings.aggregate([
    {
      $group: {
        _id: "$member_id",
        total_books: { $sum: 1 }
```

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- mongo
  - HexaDB
  - admin
  - config
  - library_db
    - members
  - local
  - retail-db
  - retail_db

```
_MONGOSH
  }
  {
    _id: 3,
    total_books: 1
  }
> db.borrowings.aggregate([
    {
      $group: {
        _id: "$book_id",
        count: { $sum: 1 }
      }
    },
    { $sort: { count: -1 } },
    { $limit: 1 }
  ])
< {
    _id: 101,
    count: 2
  }
> db.borrowings.aggregate([
    {
      $lookup: {
        from: "books",
        localField: "book_id",
        foreignField: "book_id",
```

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ata Modeling

NECTIONS (1)

arch connections

mongo

HexaDB

admin

config

library_db

members

local

retail-db

retail_db

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```
    {
      $project: {
        _id: 0,
        name: "$member.name",
        city: "$member.city",
        total_books: 1
      }
    },
    { $sort: { total_books: -1 } }
  ])
< {
    total_books: 2,
    name: 'Aarav',
    city: 'Hyderabad'
  }
  {
    total_books: 2,
    name: 'Kiran',
    city: 'Hyderabad'
  }
  {
    total_books: 1,
    name: 'Priya',
    city: 'Bangalore'
```